Study of bioinformatics

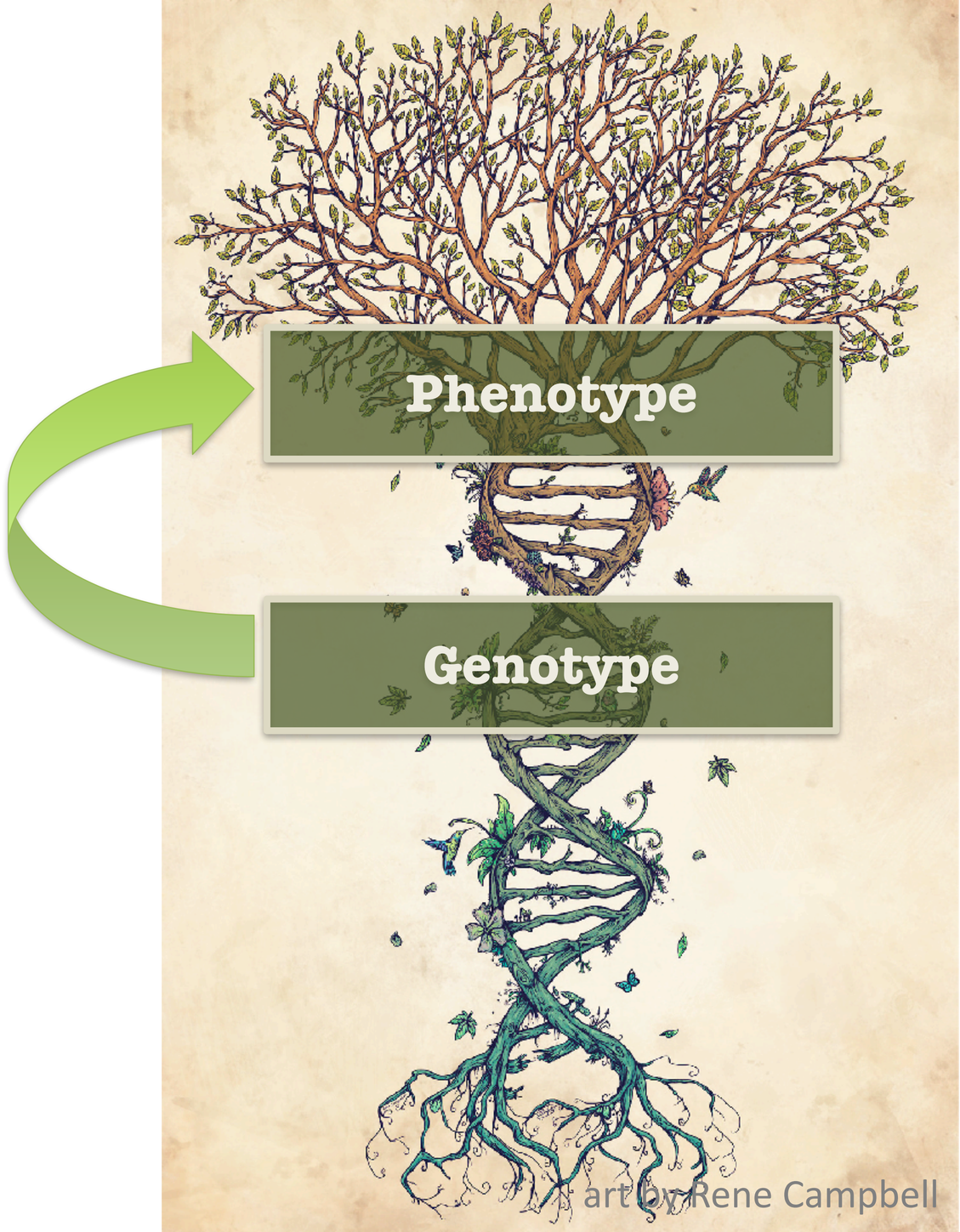
behavior

metabolism

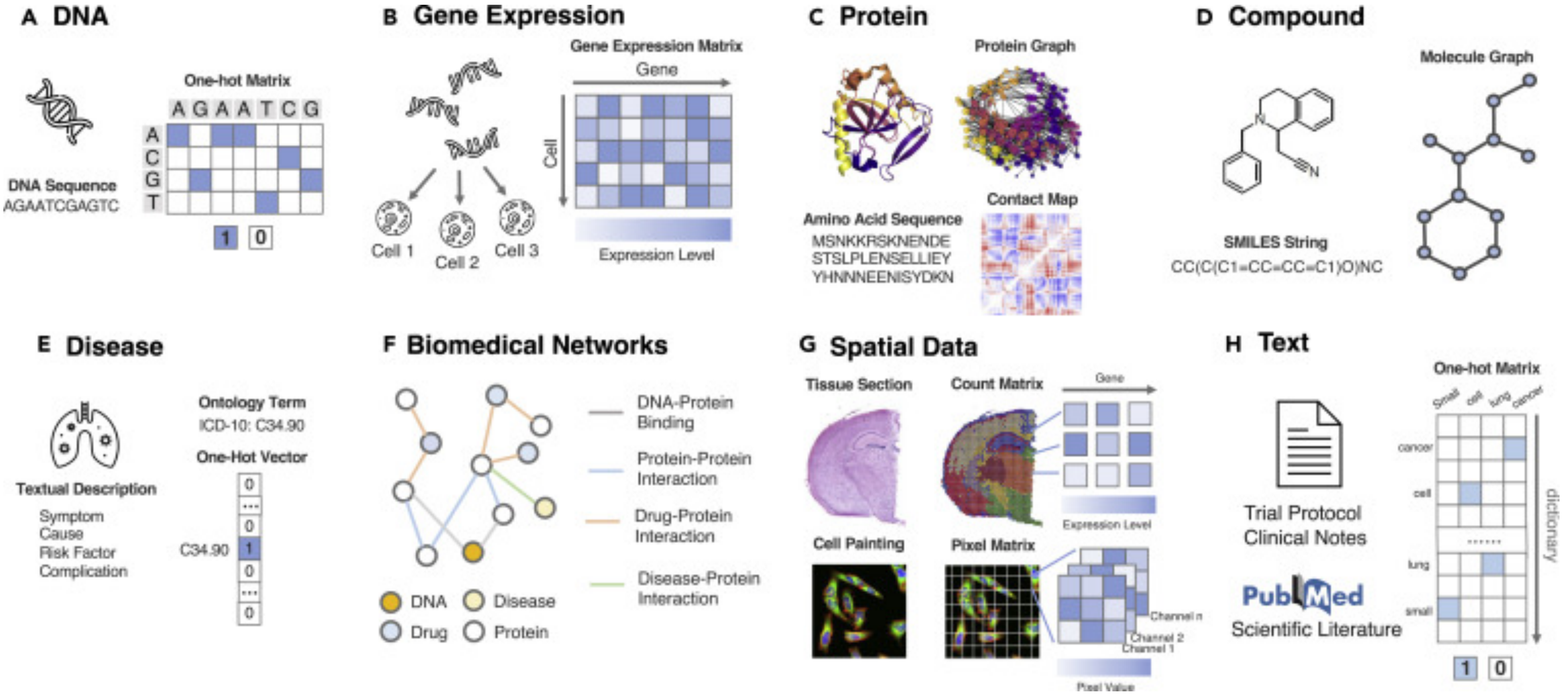
protein

RNA

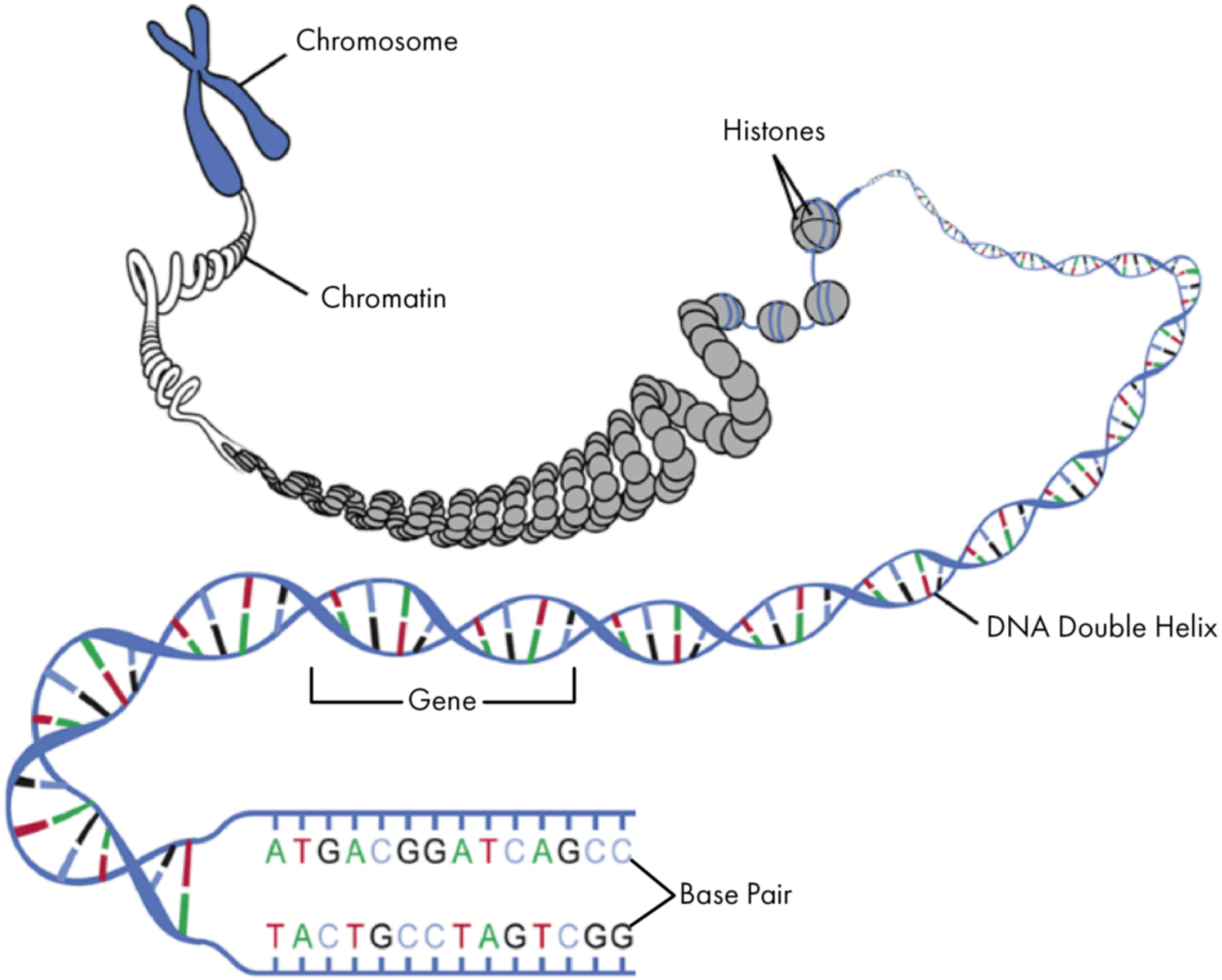
DNA



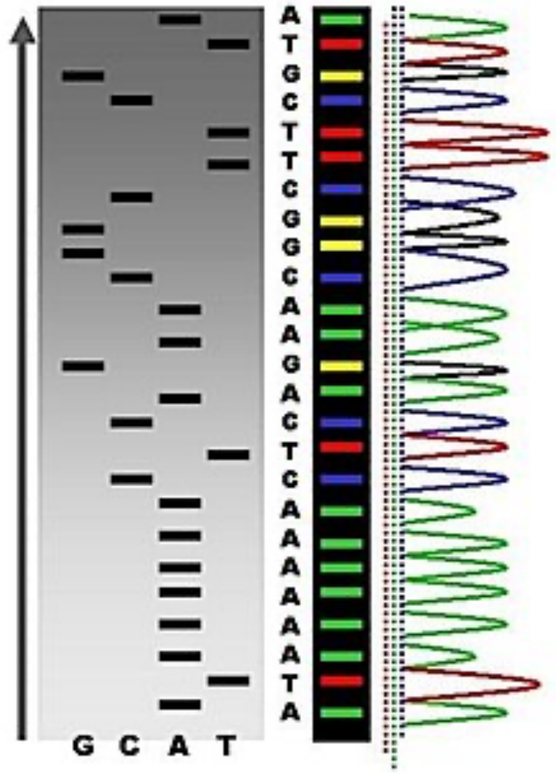
Data Types in Bioinformatics



DNA


Chromosome
Histones
Chromatin
DNA Double Helix
Gene
ATGACGGATCAGCC
TACTGCCTAGTCGG
Base Pair

Some history: Sanger Sequencing



1977: First bacterial genome
1986: Genes for color blindness
1987: Sequence of the CRISPR
1989: Gene involved in cystic fibrosis
1994: Genes involved cancer
1995: Genome of H.influenza
1996: First eukaryotic genome
1998: First multicellular genome

If you printed all your DNA
(6.4 billion letters), it would fill

4,200 books*



Veritas
sequences your
whole genome

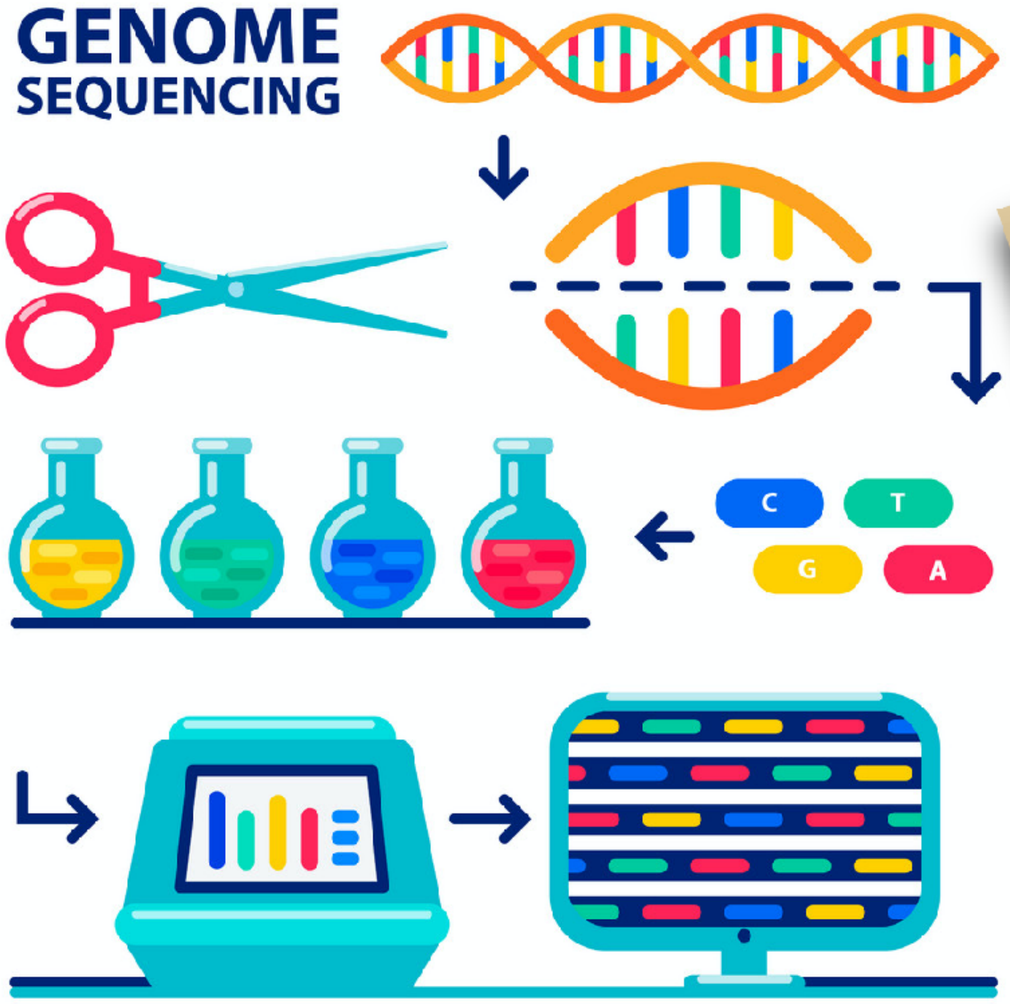
(equivalent to
4,200 books)



* Assuming that a book, like Darwin's *Origin of Species,* has 500 pages.

Human Genome Project 1990–2003

13 years
$3 billion
20 countries





What is bioinformatics?

Dictionary

Definitions from Oxford Languages · Learn more

bioinformatics

/ˌbʌɪəʊɪnfəˈmatɪks/

noun

noun: **bioinformatics**; noun: **bio-informatics**

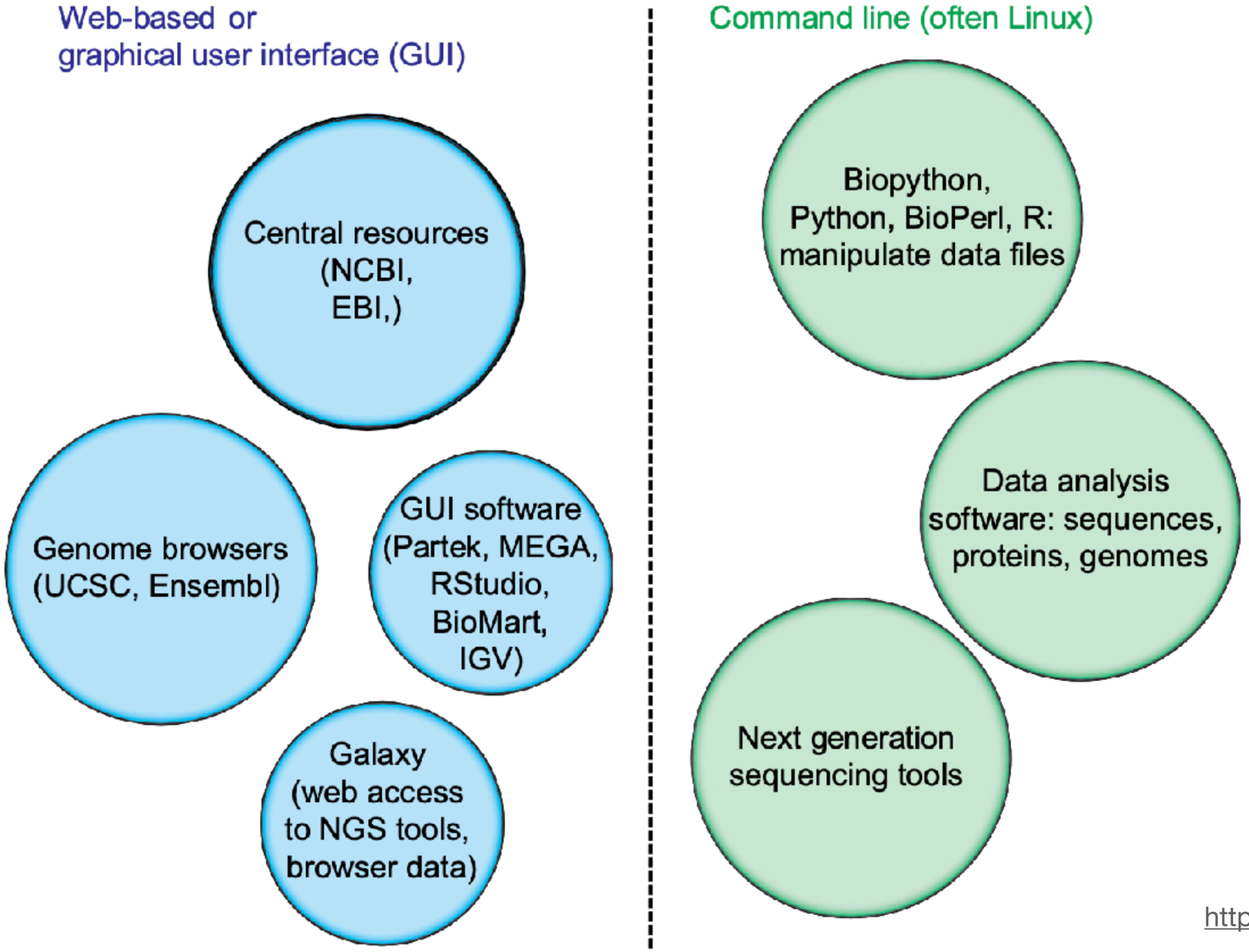
the science of collecting and analysing complex biological data such as genetic codes.

Use over time for: bioinformatics



Human Genome Project (1990-2003)

Web tools or programming?



https://opencourse.inf.ed.ac.uk/bio1

Programming



Python



R/Bioconductor



- Bioinformatics requires extensive programming skills in several languages
  - Dominated by
    - R (statistical programming) - https://cran.r-project.org, https://www.bioconductor.org
    - Python - https://www.python.org, https://biopython.org, Pandas Numpy, Scipy etc.
  - Increasing use of Machine Learning and "Data Science" methods
    - Supervised/Unsupervised Classification
    - Training of Statistical Models
    - Modelling of Data in Graph (Network) Structures
    - Bayesian Inference
    - Computational Statistics
    - Simulation
  - Ability to integrate multiple packages/tools/data sources is key

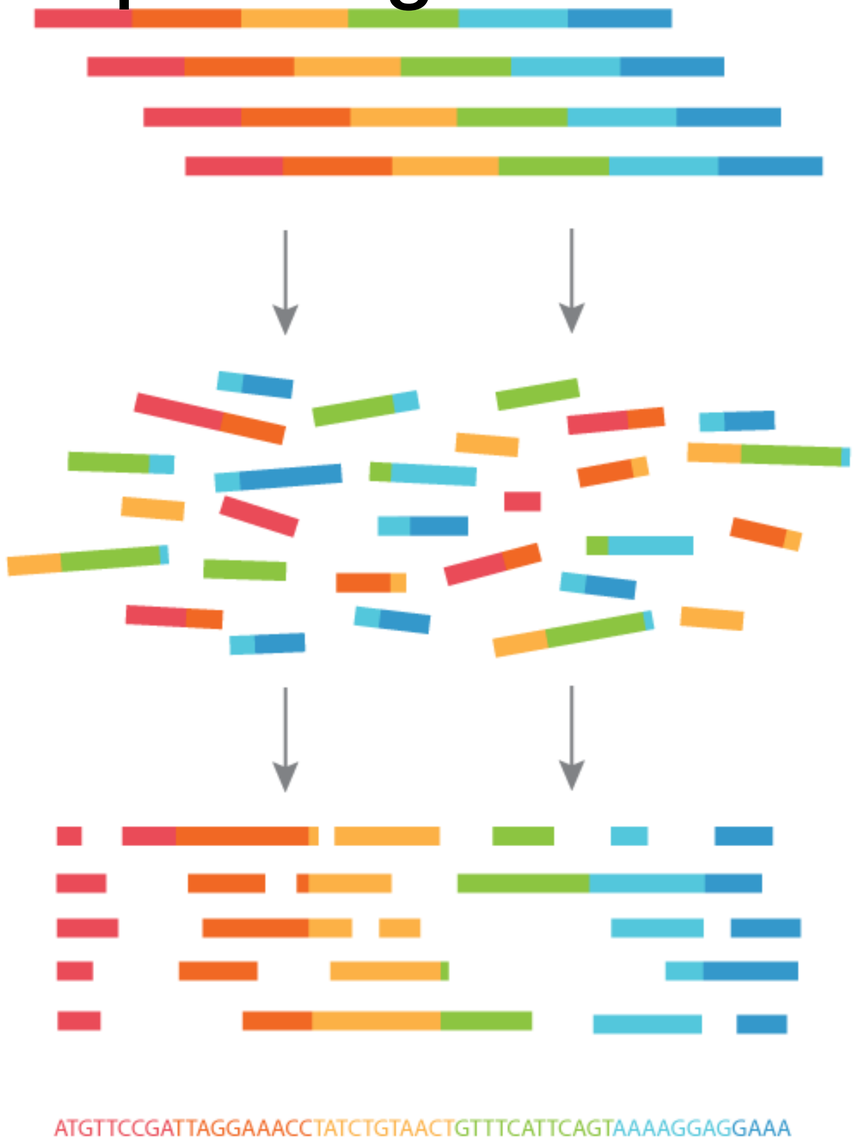
which of these have you used?

https://opencourse.inf.ed.ac.uk/bio1

A few important findings of the HGP

- ☑ Ethical breakthrough: instant data sharing
- ☑ data analysis and storage technologies
- ☑ We have ~20,000 genes
- ☑ Our chromosomes have different lengths
- ~ Most of the human genome is not coding proteins
- ~ sequence is NOT equal to function

Next Generation Sequencing



DNA Sequencing Techniques

Next Generation Sequencing

$3,000,000,000	**2003** Human Genome Project	
$20,000,000	**2006** 1st individual genome	
$2,000,000	**2007** 1st NGS Genome	
$200,000	**2008** 1st 30x genome	
$10,000	**2010** 1st sub-10K genome	
$1,000	**2014** 1st $1,000 genome	

DNA Sequencing Techniques

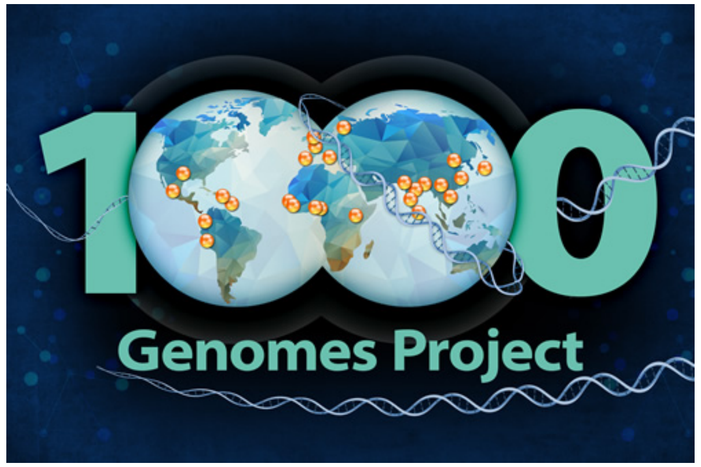
Next Generation Sequencing

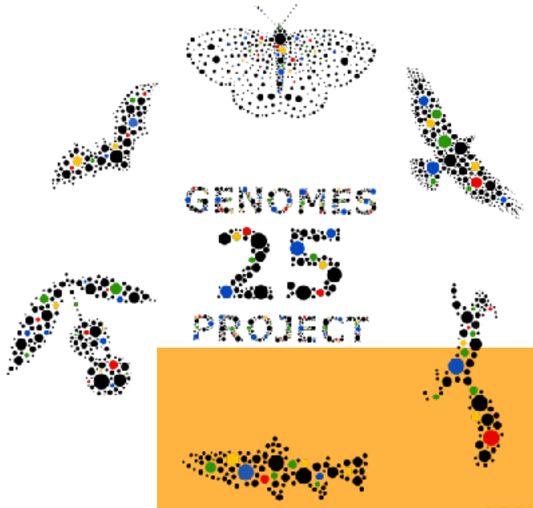
$3,000,000,000	20
$20,000,000	2006
$2,000,000	2007
$200,000	
$10,000	
$1,000	2



Genome Projects



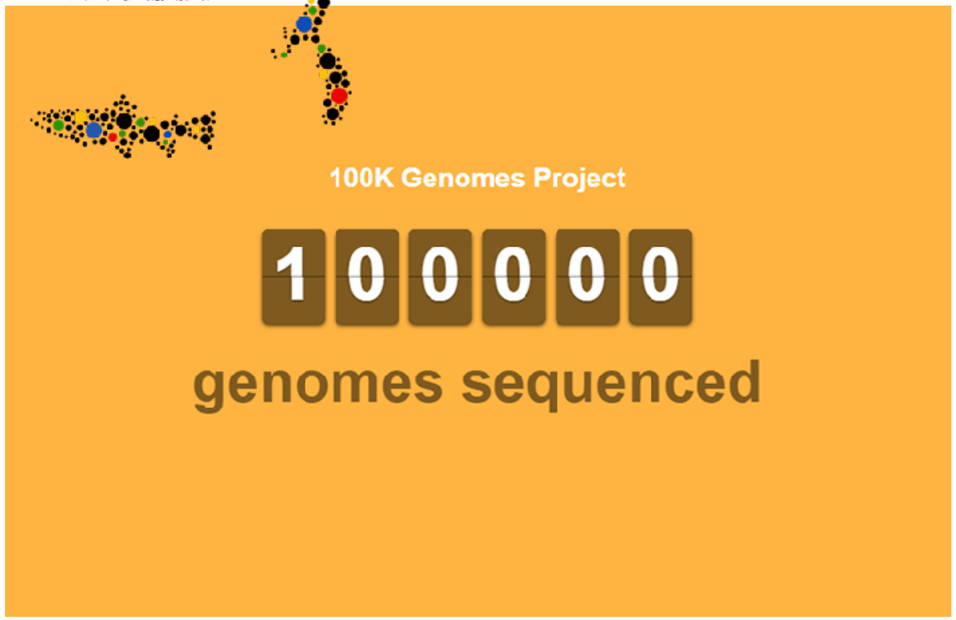




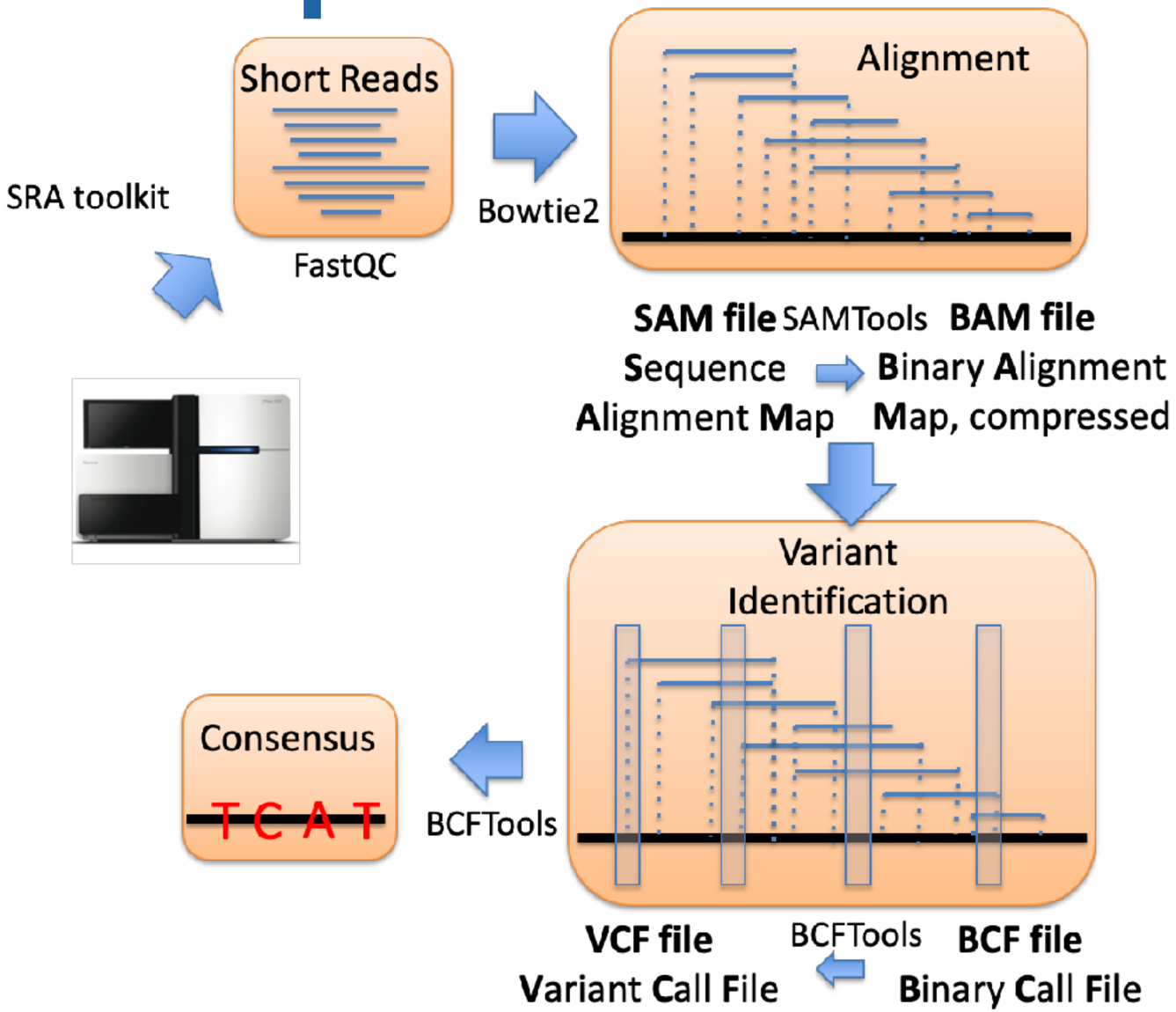


NIH HUMAN MICROBIOME PROJECT



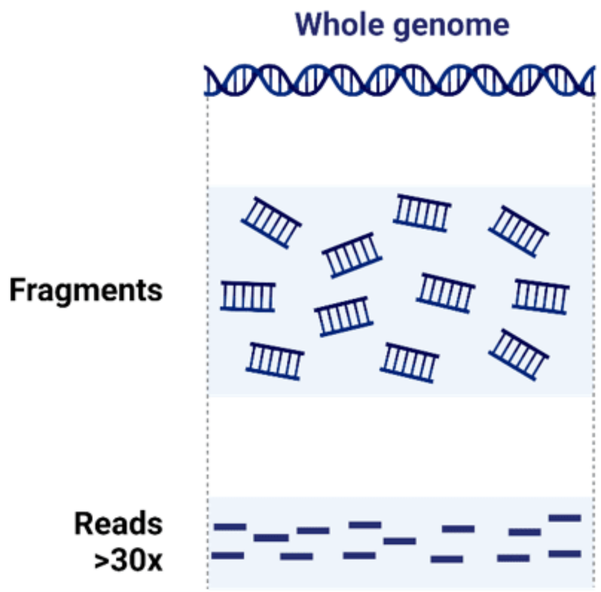


Genomics Pipeline



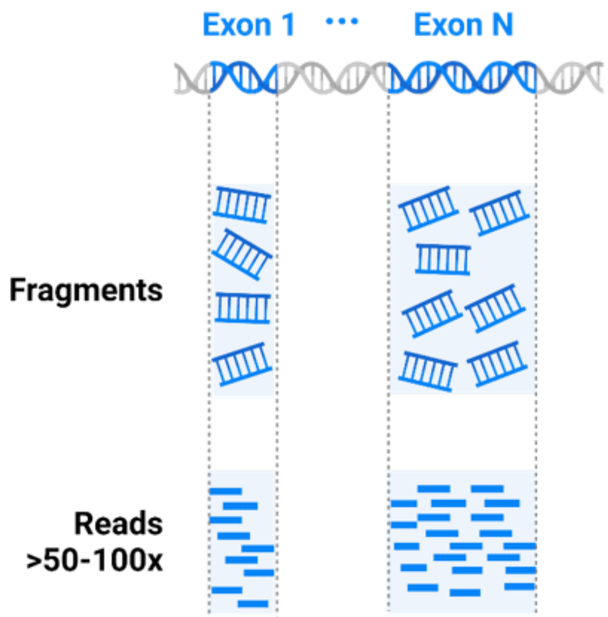
Genome, Exome, Transcriptome

WGS



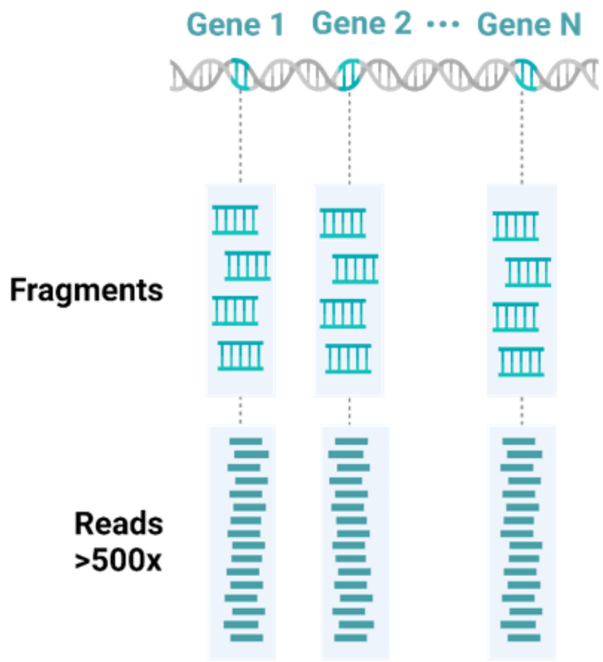
Coverage: All genes and non-coding DNA
Acuracy: Low
Variants: SNVs, InDels, SVs
Cost: Most expensive
Depth: >30X

WES



Coverage: Entire exome
Acuracy: Good
Variants: coding SNVs, InDels, CNV
Cost: Cost-effective
Depth: >50-100X

TRS

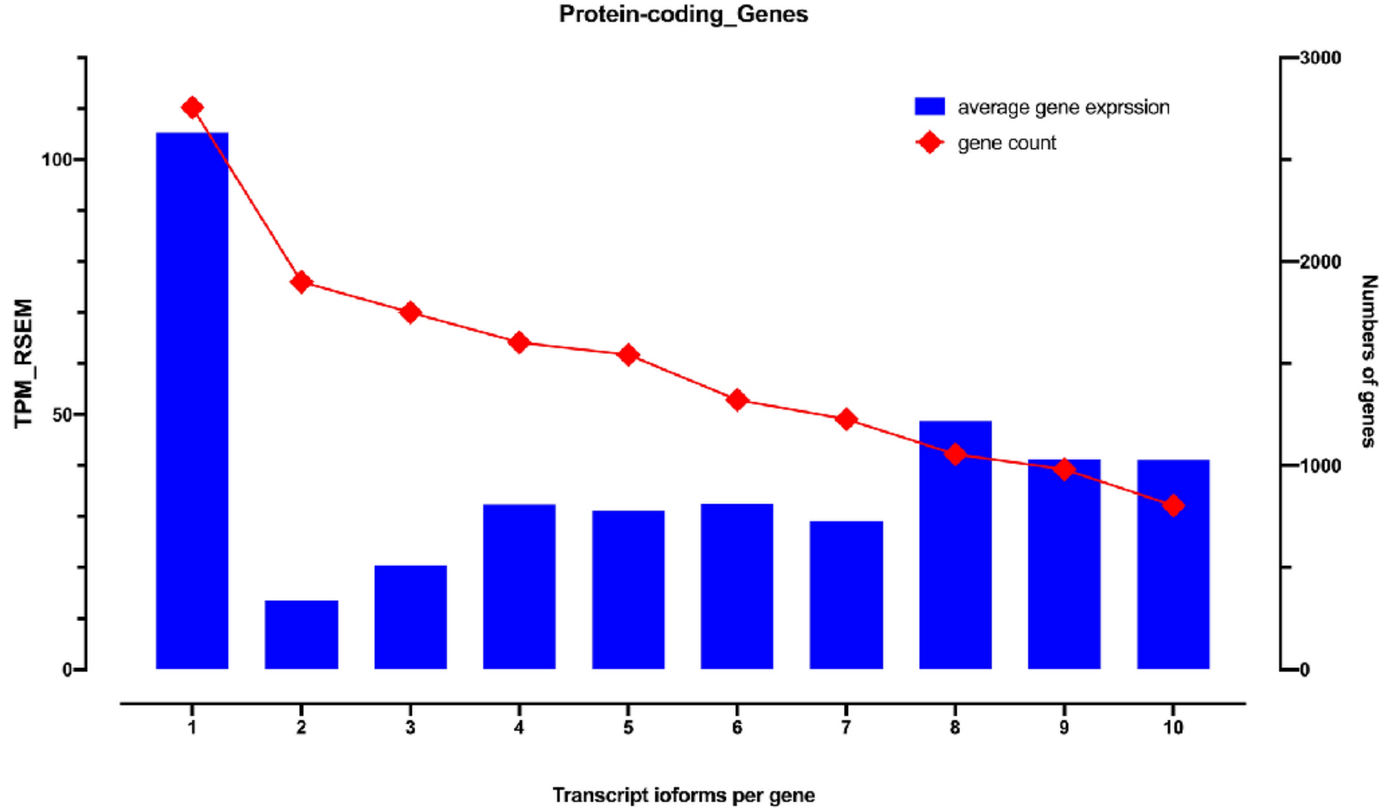


Coverage: 10-500 genes
Acuracy: High
Variants: SNVs, InDels
Cost: Most cost-effective
Depth: >500X

Novogene

Transcriptome

Although there are fewer than 22,000 protein coding genes known in the human genome, they are transcribed into over 140,000 different transcripts, over 65% of which have protein coding potential.

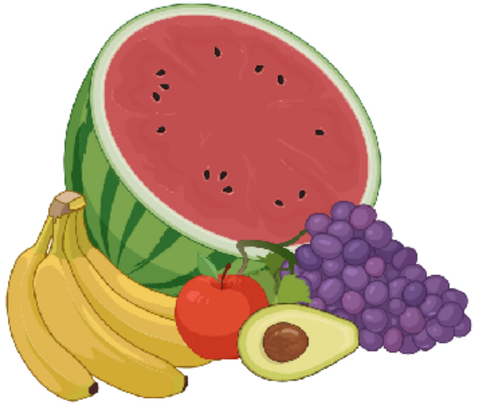


Top-ranked expressed gene transcripts of human protein-coding genes investigated with GTEx dataset

How to sequence the transcriptome?



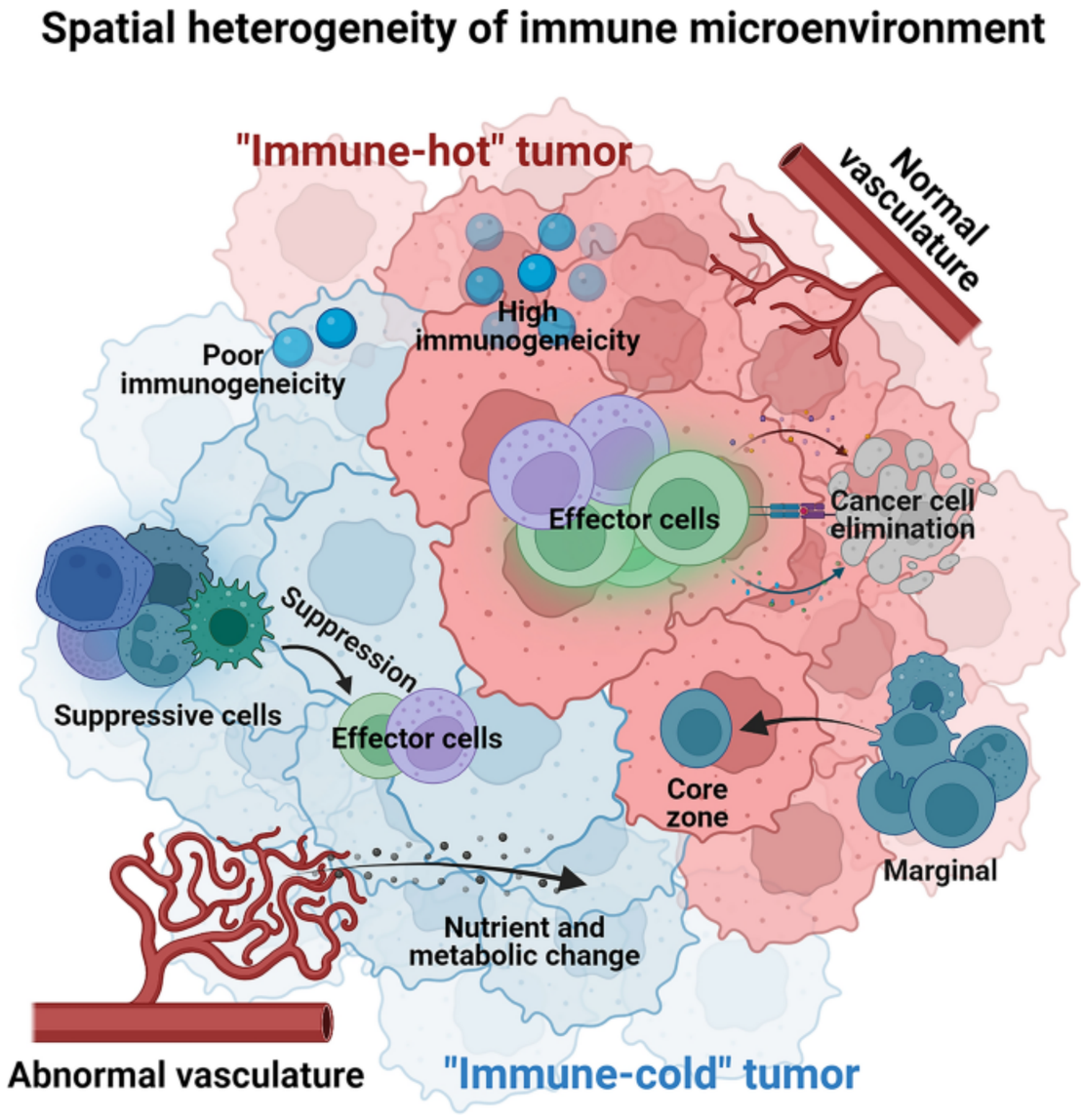
Bulk Analysis



Single Cell Analysis

BioRender

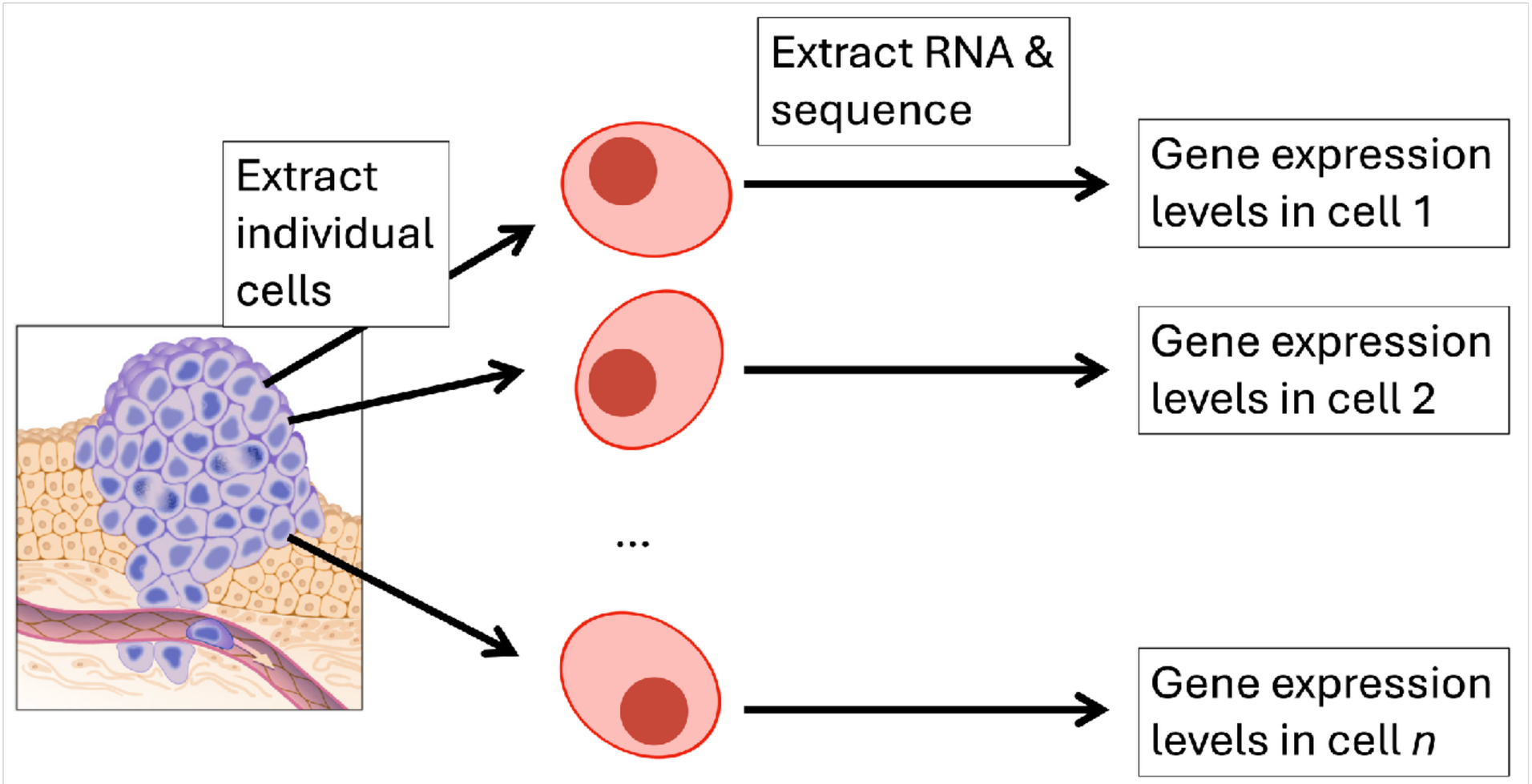
Cancer Research



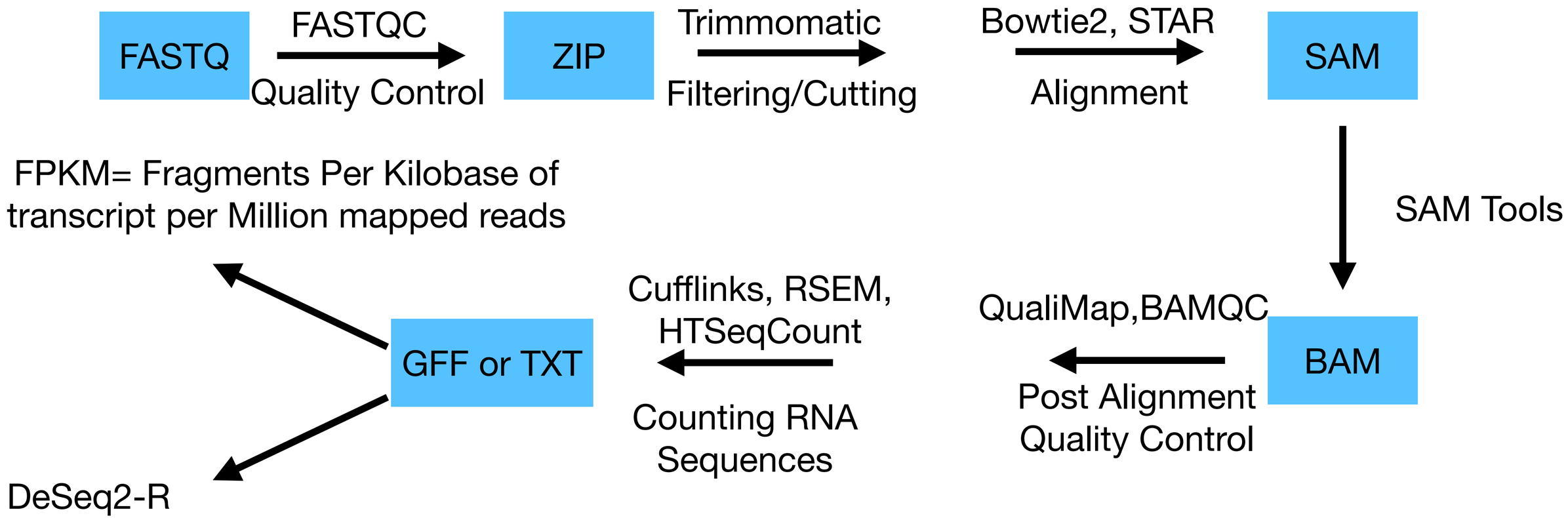
Heterogeneity of the tumor immune microenvironment and its clinical relevance

single cell Transcriptomics

In scRNAseq, individual cells are extracted from a sample of interest. RNA is extracted from each of these cells and sequenced separately. The resulting dataset now contains information about individual cells.



RNA Sequencing Analysis



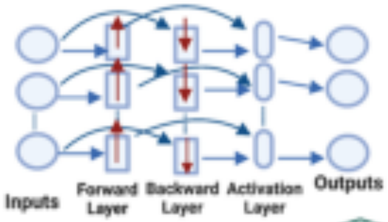
Instead of creating variant files as end products, we count the reads for example for a differential gene expression analysis

Genomics

Gene Expression & Regulation



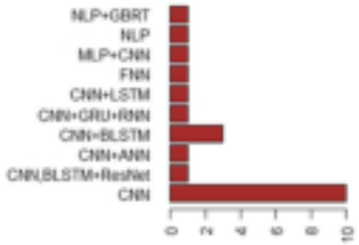
Bidirectional long short-term memory networks (BLSTM)



e.g: CLIP-seq in BED format

Predicting the effect of mutations on protein–RNA binding

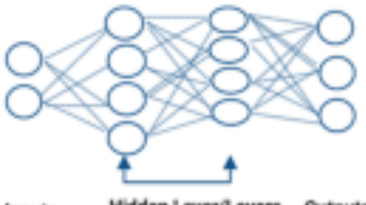
e.g. DeepCLIP



Variants calling & Annotation



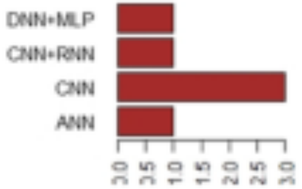
Multilayer Perceptron (MLP) or typical (ANN)



e.g: WES data in VCF

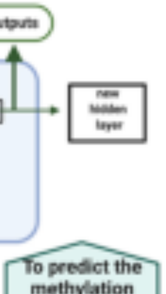
Classify true and false variants from WES data

e.g. GARFIELD - NGS



Epigenomics

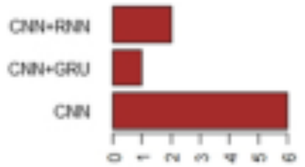
Gated Recurrent Unit (GRU)



e.g: scBS-seq or scRRBS-seq

To predict the methylation states from single-cell data

e.g. DeepCpG



Pharmacogenomic

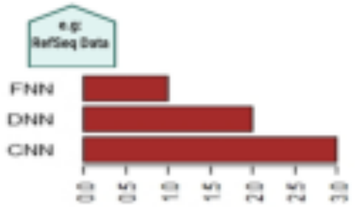
Convolutional Neural Network (CNN)



e.g: RefSeq Data

Predict the beta-lactamase (BLs) using protein or genome sequence datasets

e.g. DeepBL



Disease Variants



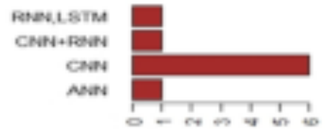
Recurrent Neural Network (RNN)



e.g: ChIP-seq

Predict non-coding seq. variants impact

e.g. DeepMILO



Mutations

- **Single Nucleotide Polymorphisms – SNPs**

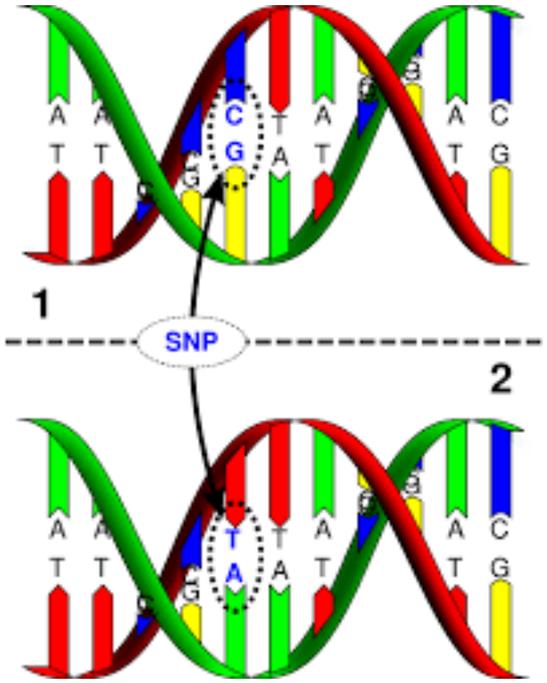
Mutation in 10-100 million bases per generation per cell

- **Indels**

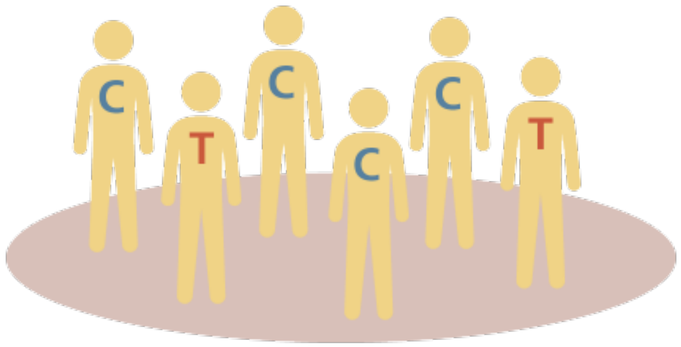
Insertions and deletions of sequences, less frequent

- **Copy Number Mutations**

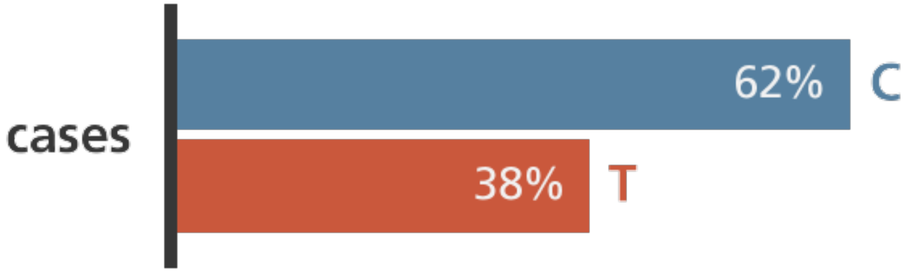
Long or short sequences

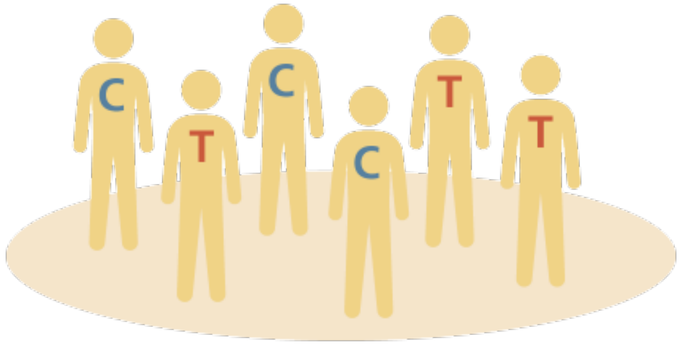


Genome Wide Association Studies



cases (n=1,000)
people with heart disease





controls (n=1,000)
people without heart disease

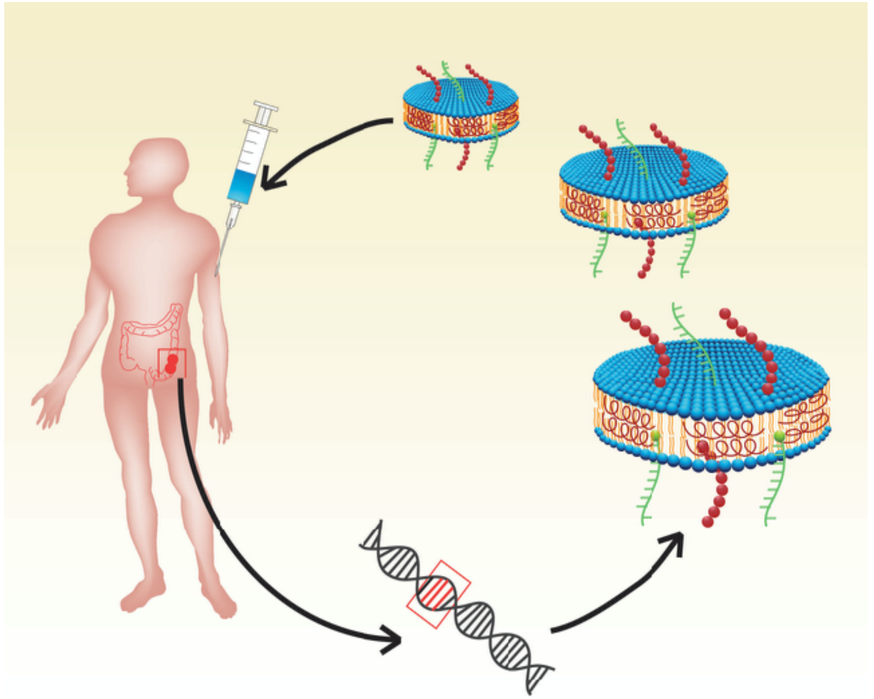


Cancer Genomics and Precision Medicine

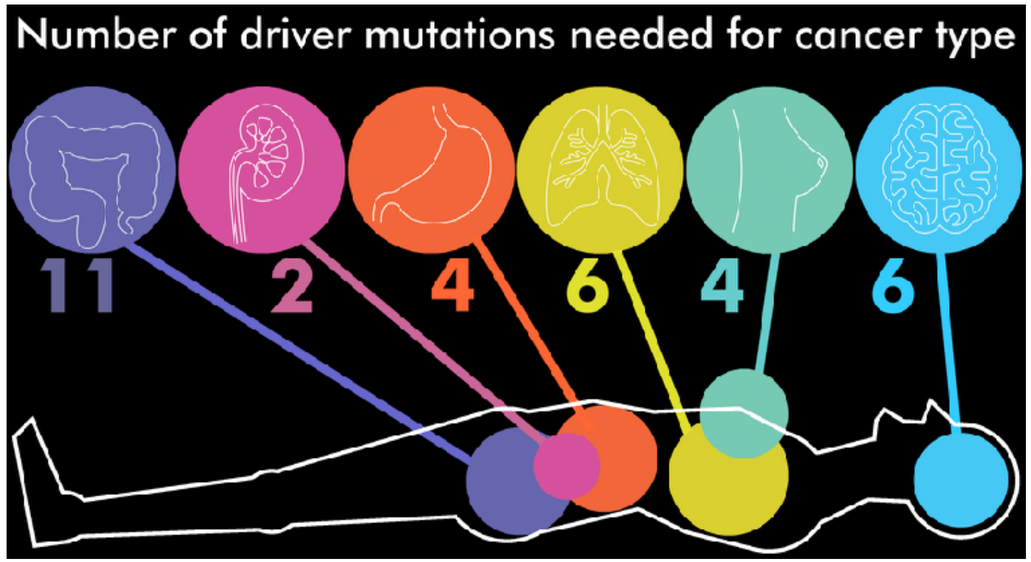
Matched tumor & normal tissues from more than **11,000** patients, representing **33** cancer types.





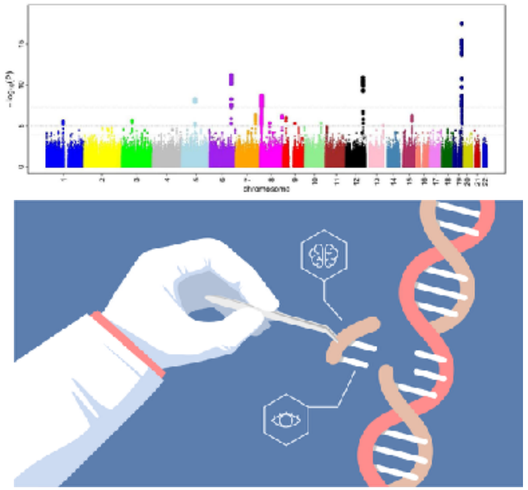


Most mutations are neutral

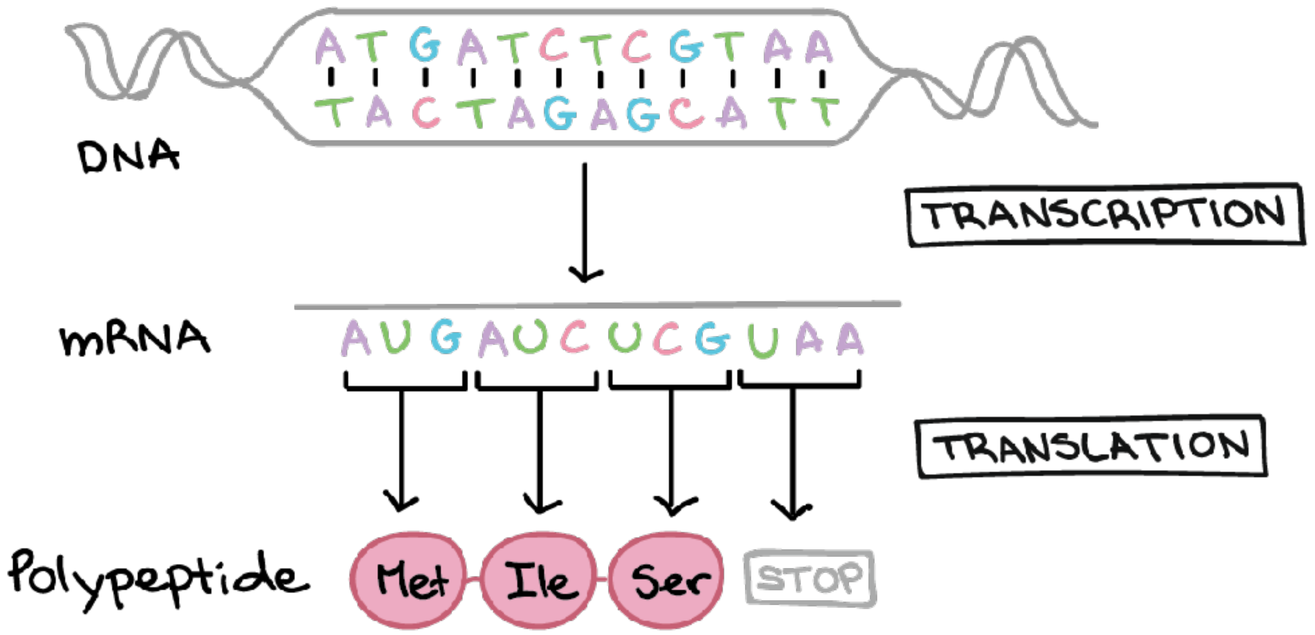


Which mutations?

- In silico candidates - statistical prediction
- In vitro mutagenesis - experimentally mutate

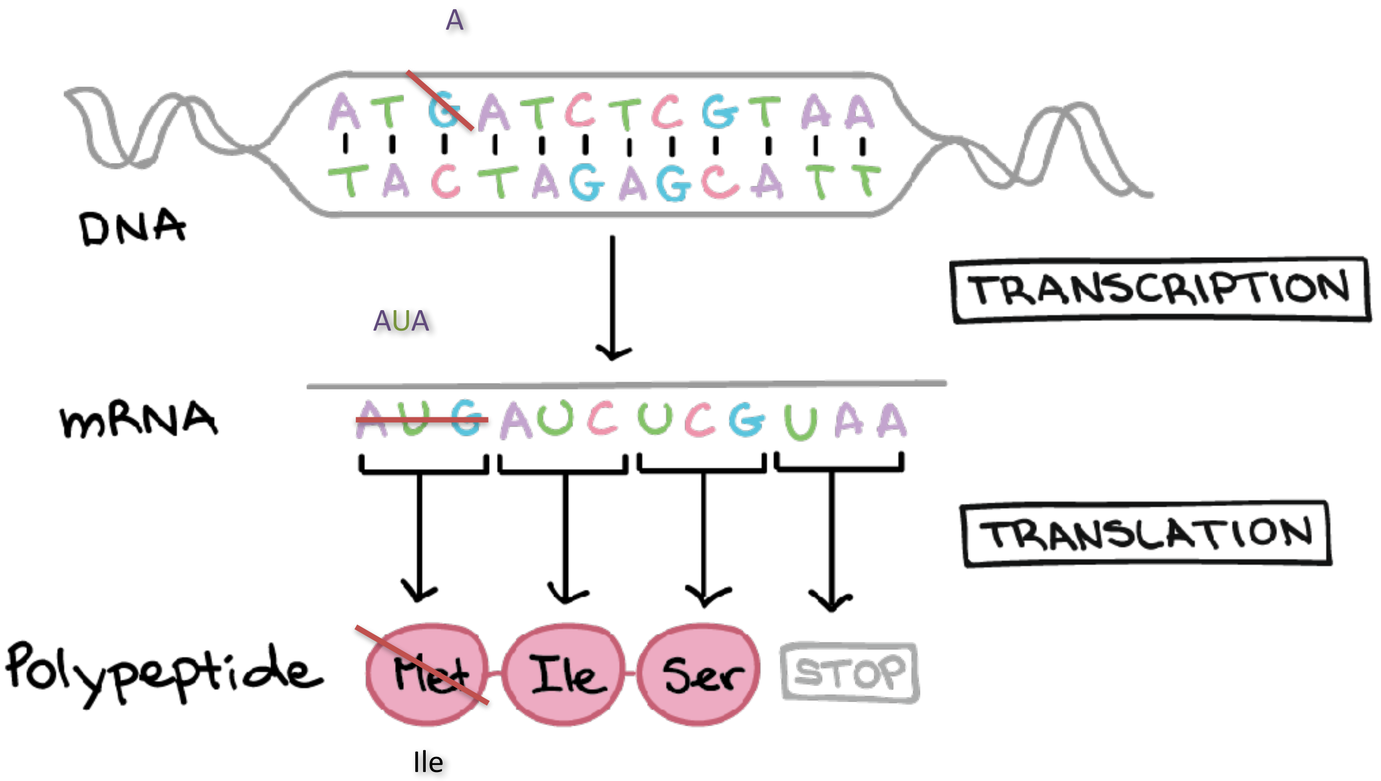


THE CENTRAL DOGMA



GCA GCC GCG GCU	AGA AGG CGA CGC CGG CGU	GAC GAU	AAC AAU	UGC UGU	GAA GAG	CAA CAG	GGA GGC GGG GGU	CAC CAU	AUA AUC AUU	UUA UUG CUA CUC CUG CUU	AAA AAG	AUG	UUC UUU	CCA CCC CCG CCU	AGC AGU UCA UCC UCG UCU	ACA ACC ACG ACU	UGG	UAC UAU	GUA GUC GUG GUU	UAA UAG UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

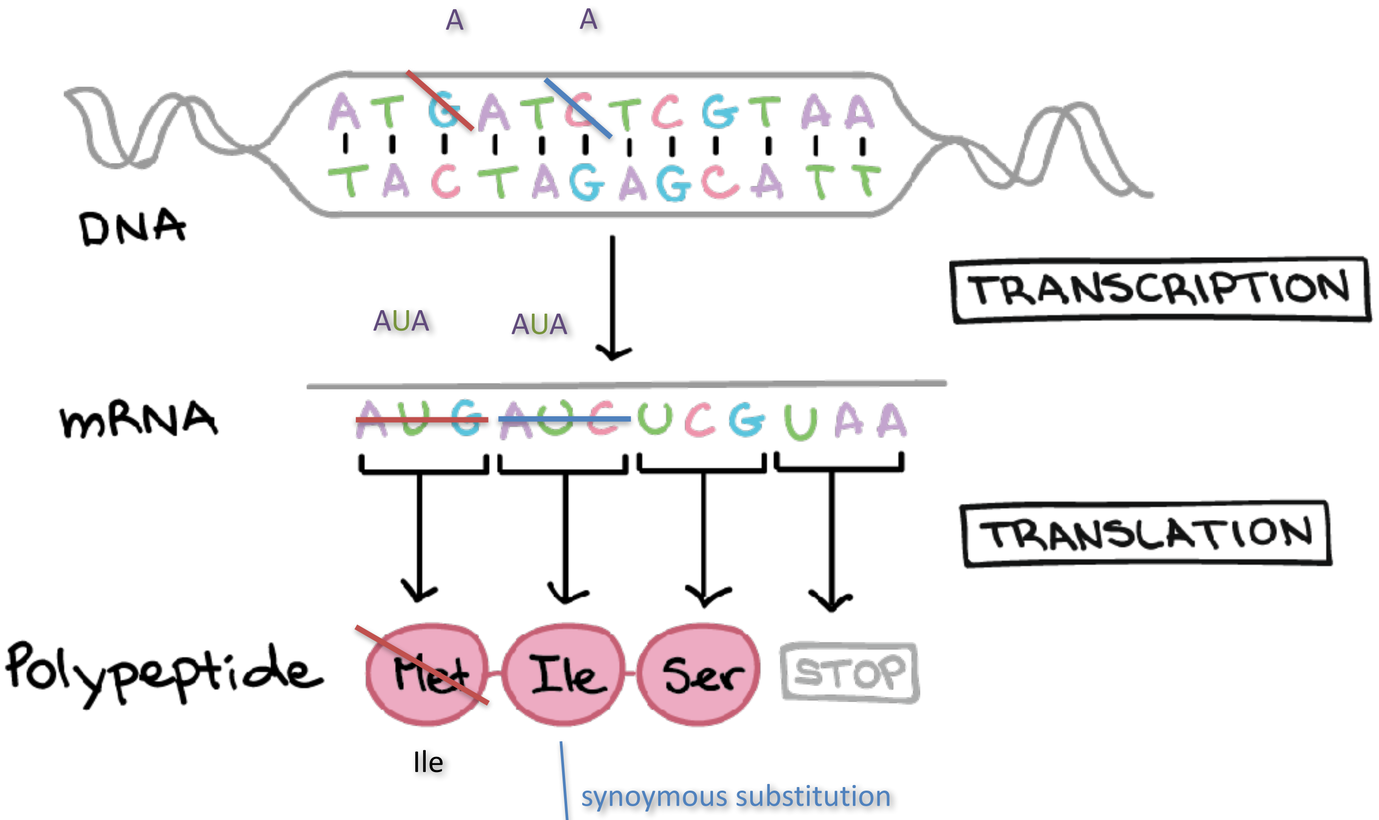
THE CENTRAL DOGMA



nonsynoymous substitution

GCA GCC GCG GCU	AGA AGG CGA CGC CGG CGU	GAC GAU	AAC AAU	UGC UGU	GAA GAG	CAA CAG	GGA GGC GGG GGU	CAC CAU	AUA AUC AUU	UUA UUG CUA CUC CUG CUU	AAA AAG	AUG	UUC UUU	CCA CCC CCG CCU	AGC AGU UCA UCC UCG UCU	ACA ACC ACG ACU	UGG	UAC UAU	GUA GUC GUG GUU	UAA UAG UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

THE CENTRAL DOGMA



GCA GCC GCG GCU	AGA AGG CGA CGC CGG CGU	GAC GAU	AAC AAU	UGC UGU	GAA GAG	CAA CAG	GGA GGC GGG GGU	CAC CAU	AUA AUC AUU	UUA UUG CUA CUC CUG CUU	AAA AAG	AUG	UUC UUU	CCA CCC CCG CCU	AGC AGU UCA UCC UCG UCU	ACA ACC ACG ACU	UGG	UAC UAU	GUA GUC GUG GUU	UAA UAG UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

Identifying positive selection

Nonsynonymous / Synonymous substitution **>> 1** ➡ **positive selection**

TCCGATATATGGCAACCCGACAAA
S D I W Q P D K

TCAGATCTATGGCAGCCCCACAAA
S D L W Q P R K

Codon Models

codon substitutions over evolutionary time





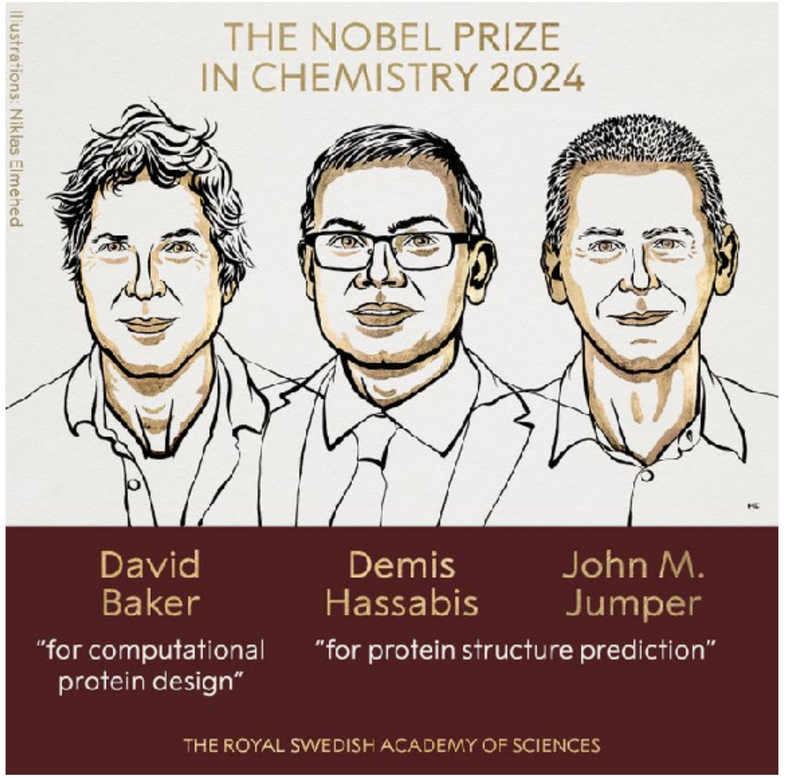
CodonPhyML: Fast Maximum Likelihood Phylogeny Estimation under Codon Substitution Models

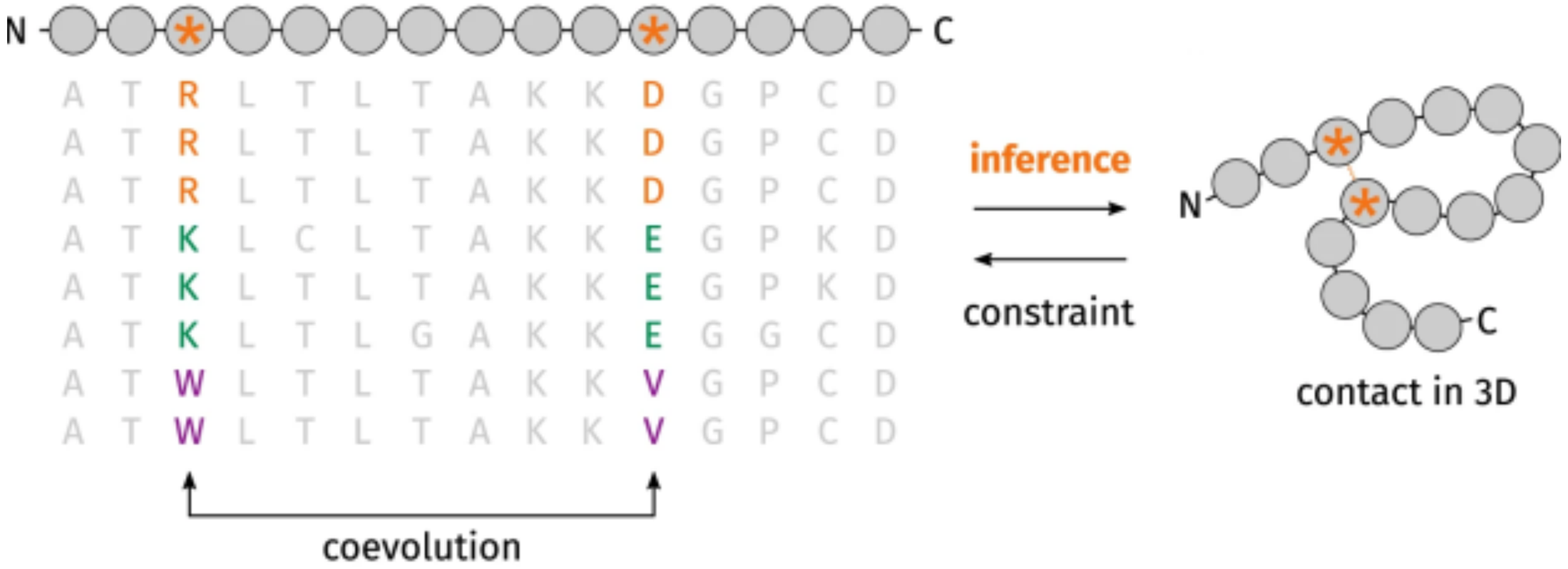
Manuel Gil,[*,†,1,2,3] Marcelo Serrano Zanetti,[†,1,4] Stefan Zoller,[1,2] and Maria Anisimova[*,1,2]

- account for variable substitution rates
- use phylogeny
- incorporated variation of selective pressure among sites along a gene/protein and different branches of the tree
- use maximum likelihood approaches to test for positive selection

AI uses sequence alignment

and how it initiated AlphaFold to help us predict structure from sequence





Here's a paper I just found: https://www.nature.com/articles/s41592-024-02201-0. Can you summarize it in one sentence and tell me how many data points they trained on?

Language model

The paper introduces scGPT, a foundation model for single-cell biology created by leveraging generative pretrained transformers on over 33 million cells' data, aimed at advancing genetic and cellular biology research by distilling critical insights from single-cell sequencing data .

Write code using Scanpy to load an example single-cell gene-expression dataset and visualize it using UMAP

Language model

```
import scanpy as sc

# Load the PBMC dataset
adata = sc.datasets.pbmc3k()

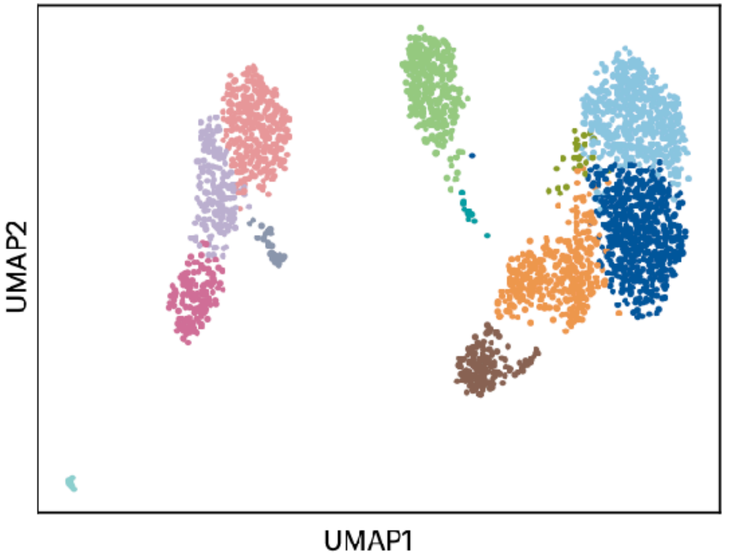
# Preprocess the data
sc.pp.normalize_total(adata)
sc.pp.log1p(adata)
sc.pp.pca(adata)
sc.pp.neighbors(adata)

# Cluster the cells
sc.tl.leiden(adata)

# Perform UMAP dimensionality reduction
sc.tl.umap(adata)

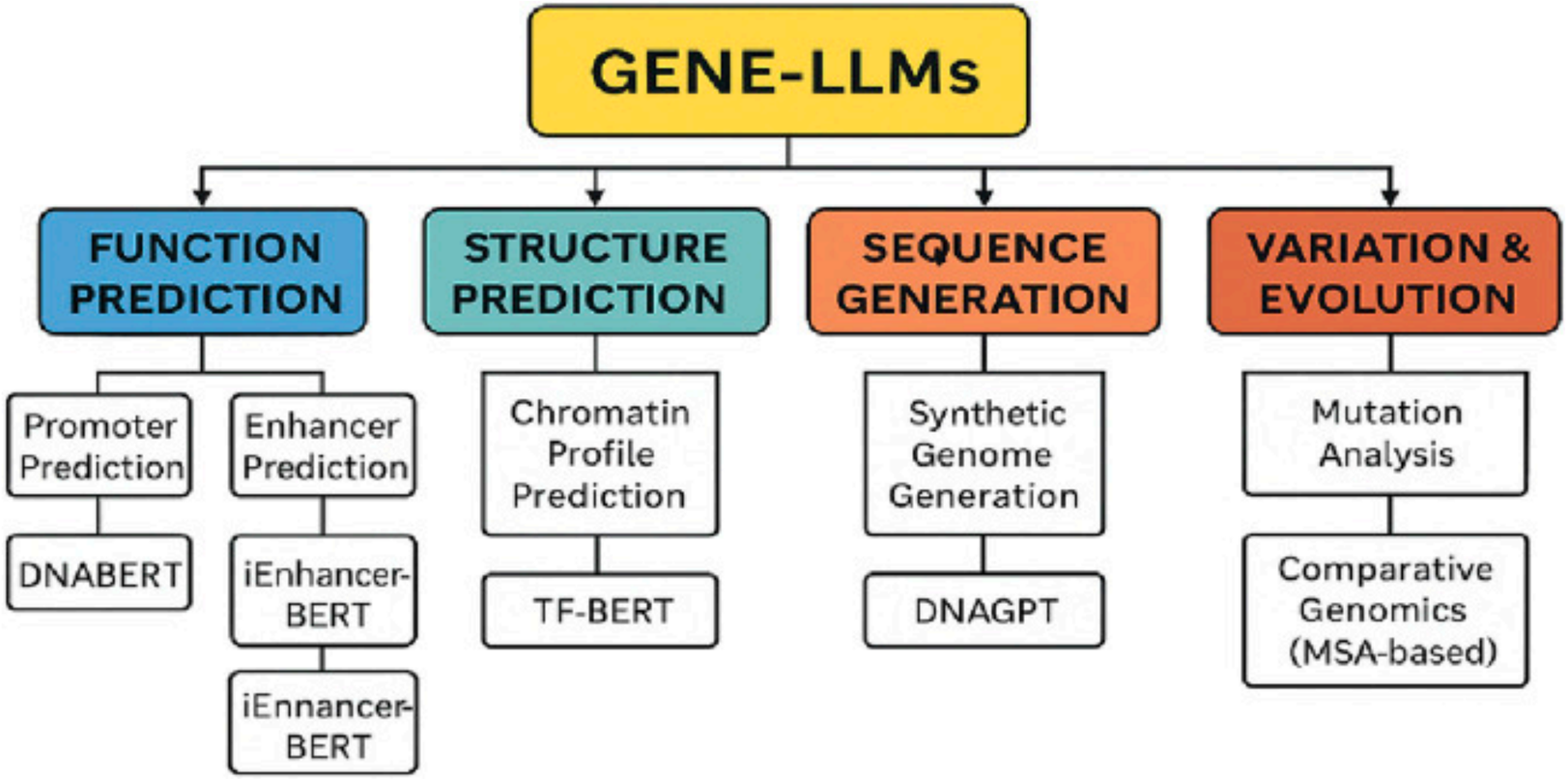
# Visualize the data
sc.pl.umap(adata, color=['leiden'])
```

Python



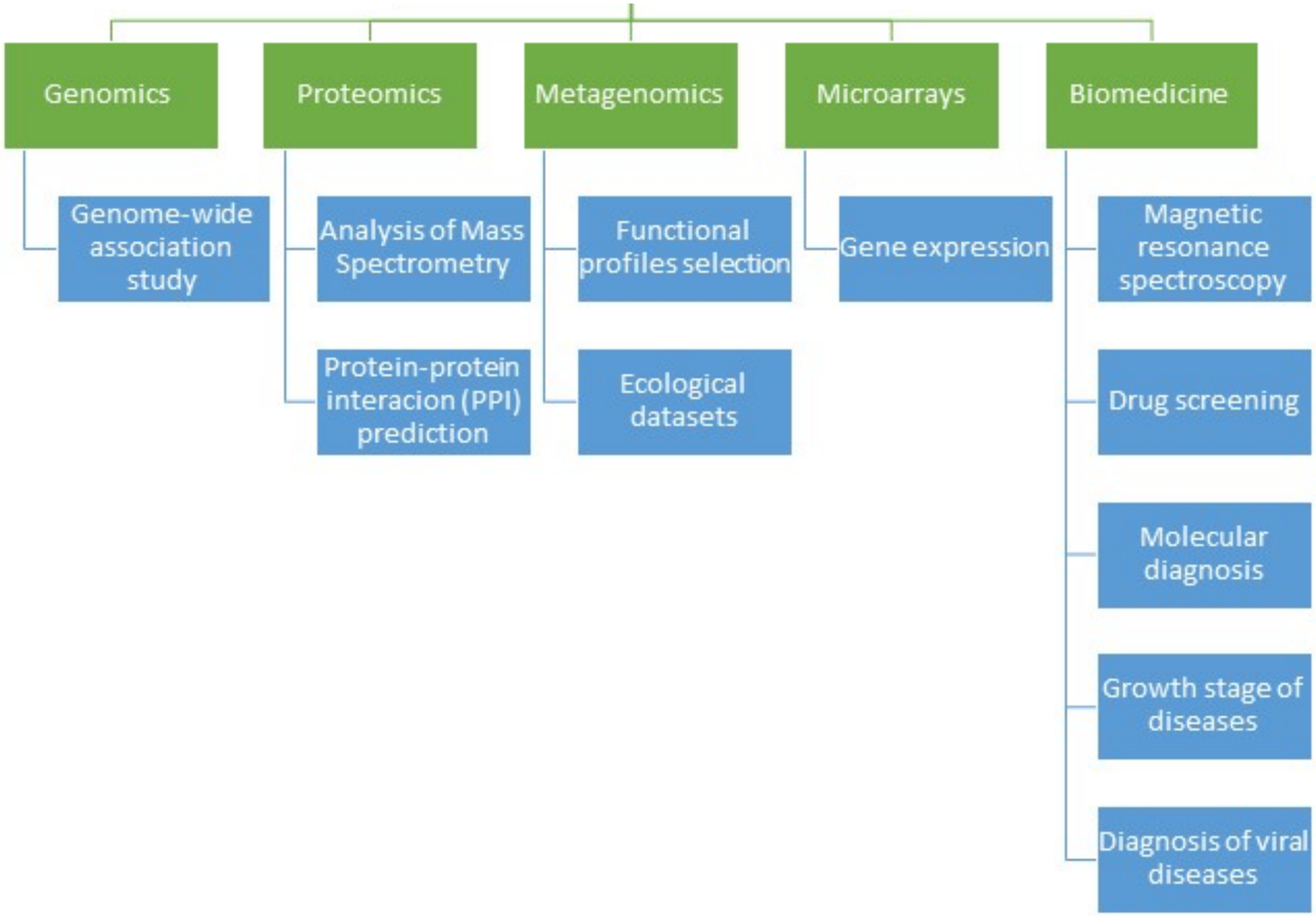
Language models for biological research: a primer, Nature, 2024

genomic LLMs



Machine learning uses in bioinformatics

- Artificial neural networks
- PCA, tSNE etc.
- Classification
- Convolutional neural networks
- Random forest



Protein Databases

- Protein-protein interaction: https://string-db.org/
- Protein databases: https://www.uniprot.org/ https://www.rcsb.org/

Bioinformatics Forums

- Best friend of a bioinformatician: https://www.biostars.org
- Another best friend: https://stackoverflow.com

Notebooks

- Jupyter Notebook, bioinformatician's diary: https://jupyter.org

Downstream Genomic

- Bedtools : https://bedtools.readthedocs.io/en/latest/

Omics Databases

- USCS Table Browser: https://genome.ucsc.edu/cgi-bin/hgTables
- Biomart Ensembl: https://useast.ensembl.org/index.html
- ncbi entrez: https://www.ncbi.nlm.nih.gov/

Cancer Databases

- Pan Cancer, TCGA: https://cancergenome.nih.gov

Gene Databases

- Gene Cards: https://www.genecards.org
- DAVID for Gene Ontology terms: https://david.ncifcrf.gov/summary.jsp

Genome Databases

1000 genomes new name : https://www.internationalgenome.org

Coding

- R Circos - https://cran.r-project.org/web/packages/BioCircos/vignettes/BioCi
- R packages: https://www.bioconductor.org
- python packages: Biopython